

Post-Submission Experiments: From the Diagnostic Task to Controllable Generation

Experiment record, generated during defense preparation

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Abstract

The submitted thesis diagnoses why an input-embedding gradient of a frozen autoregressive likelihood carries no usable directional signal, and demonstrates a repair, the relaxed token-indicator (one-hot) surrogate, on a single-token recovery task. It never crosses the bridge to controllable generation. This document records the experiments run after submission to cross that bridge, or to measure exactly why it cannot be crossed. Four findings are load-bearing. First, the repair does not need a gradient at all: the self term $\log p(v \mid \mathbf{x}_{<i})$, which the forward pass already computes, matches the full token-indicator surrogate at one quarter of the cost, with zero backward passes. Second, the thesis’s headline 40% recovery figure reproduces exactly at its own $n = 50$ but settles at 26.5% at $n = 200$, which removes the claimed parity with a purpose-trained diffusion model while leaving the zero-versus-nonzero finding untouched. Third, implementing MuCoLa’s actual energy in the same code makes the continuous sampler produce fluent text where the thesis’s energy produces word salad, which confirms the scope restriction the submitted thesis already states. Fourth, a likelihood-ranked shortlist and a sentiment constraint want almost unrelated tokens, and the classifier’s own first-order expansion has a rank correlation near zero with the change it is expanding, which supplies a mechanism for the weakness of gradient-based constraint guidance in this setting.

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1 Purpose, scope and what is authoritative

The thesis is submitted and nothing in it is changed by this work. This document exists so that the author can answer, with numbers, the question the thesis leaves open: is the sampler actually useful for controllable generation?

Four questions were posed, in priority order.

- Q1 Cost.** The output-side surrogate decomposes into a *self* term, $\log p(v \mid \mathbf{x}_{<i})$, which the energy’s own forward pass already computes and which is therefore free, and a *future* term describing how substituting v changes everything downstream, which costs either one backward pass (linearized) or k forward passes (exact). Is the future term worth its cost?
- Q2 Conditioning.** Do bidirectional proposals, which see right context, beat the autoregressive output-side proposal when the controlled attribute is global?
- Q3 Mechanism.** How should a constraint enter: exact evaluation on a shortlist, a first-order surrogate of the classifier, both, or only the accept step?
- Q4 Usefulness.** At matched compute, does the exact-energy Metropolis–Hastings chain beat MuCoLa on the adherence-fluency trade-off, and does it produce more distinct valid solutions?

The authoritative machine-readable record is `CTG_EXPERIMENTS_LOG.md` in the repository root. Every number in this document carries the path of the JSON it came from. Suggested defense slides are in `SUGGESTED_BACKUP_SLIDES.md` and are suggestions only; nothing has been inserted into the deck.

1.1 Relationship to the submitted documents

This file lives in `Doc/final/future_work/` and is a new addition. It does not modify, include or depend on `Doc/final/thesis/`, `Doc/final/proposal/` or `Doc/final/beamer/`. Those three directories are byte-for-byte unchanged from the submitted state; their checksums are recorded in the repository log.

2 Part 0: audit of what already existed

2.1 The constrained task was reused, not rebuilt

`scripts/run_constrained.py` already defines the whole constrained-generation task, and it was imported rather than reimplemented: the fifteen PPLM discriminator prompts verbatim from MuCoLa’s repository, the twenty-token continuation span, the centroid and random-token initializations, the two setups, the five constraint-mode arms, and the sentiment head. Only the sampler and the recording are new.

2.2 What `policy_onehot` actually computes

This determined the arm list and was established by reading the code rather than from memory. In `core/dls.py` the proposal logit for candidate v is built as

$$t_2[v] = \underbrace{\frac{1}{2} \mathbf{g}^\top (\mathbf{e}(v) - \mathbf{s})}_{\text{embedding-gradient (future) term}} + \underbrace{\frac{1}{2} \log p(v \mid \mathbf{x}_{<m})}_{\text{self term, added only for policy_onehot}}$$

so `policy_onehot` is the self term *plus* the input-embedding-gradient term, not the self term alone. It costs one forward, one backward and one extra no-grad forward per step, and the same again for the Metropolis reverse evaluation.

Consequently the thesis’s headline one-hot result had never been decomposed into its two halves. Doing so is the first experiment below, and it is cheap.

3 Study A: is the future term worth its cost?

3.1 Arms and pre-registration

Arm	Proposal score for candidate v	Model calls per step
A1 <code>policy_self</code>	$\frac{1}{2} \log p(v \mid \mathbf{x}_{<m})$	1 fwd, 0 bwd
A2 <code>policy_onehot</code>	$A1 + \frac{1}{2} \mathbf{g}^\top (\mathbf{e}(v) - \mathbf{s})$	2 fwd, 1 bwd
A3 <code>policy_exact_k</code>	$\frac{1}{2} (E(v) - E(\text{cur}))$ on a shortlist	1 fwd + k fwd
A4 <code>policy</code>	$\frac{1}{2} \mathbf{g}^\top (\mathbf{e}(v) - \mathbf{s})$ alone	1 fwd, 1 bwd
A5 <code>uniform</code>	constant	1 fwd, 0 bwd

Table 1: Study A arms, all inside the exact-energy Metropolis–Hastings chain.

Three predictions were recorded in advance: (1) A1 approximately equals A2 at strictly lower cost; (2) A3 buys little over A1 relative to its cost; (3) A4 stays at the floor with A5.

3.2 Two framings

Framing 1, full vocabulary. Every arm proposes over all 50,257 tokens with the Langevin distance term present, in the cell that produced the thesis’s headline one-hot result ($\epsilon : 10.5 \rightarrow 0.1$, $T = 1.0$, Metropolis on, gradient normalization off). Directly comparable to the published figures. A3 cannot appear here: exact scoring of the full vocabulary would cost 50,257 forward passes per step.

Framing 2, matched shortlist. Every arm proposes over the same frozen top- k candidate set ($k \in \{16, 64, 256\}$) scored only by its own surrogate, distance term dropped. This is the clean comparison, because the exact arm can only ever see a shortlist, so the restriction is charged to every arm equally.

3.3 Three defects found before any real run

Each of these would have produced a confident and wrong answer, and each was caught by a smoke test.

1. **Shortlist excludes the incumbent.** The exact arm put zero mass on the token the chain currently held, so the reverse probability was zero, $\log \alpha$ was $-\infty$, and acceptance was 0.0% on every sequence.
2. **The distance term pins a shortlisted chain.** With $k = 64$ and the Langevin term t_1 present, all five arms returned an identical final KL of 6.777 and zero exact matches, because t_1 scores the incumbent at exactly 0 and every rival at $-\|e(v) - s\|^2/2\epsilon$. Over 50k candidates that is one preference among many; over 65 it is decisive.
3. **A state-dependent shortlist is not reversible.** A shortlist recomputed at each state as “top- k here plus wherever I am” gives the reverse move probability zero as soon as the chain leaves its starting token, which it does immediately. Acceptance was again 0.0% while the proposal itself was healthy: measured directly, 24.6% of the proposal mass sat on the ground-truth token and the move was worth +19.8 nats, and it was refused because the reverse term was $-\infty$. Fixed by freezing the support per sequence, which makes each arm an independence sampler on a fixed candidate set, a valid Metropolis–Hastings kernel.

A regression gate was run after every sampler edit: the flagship archived configuration reproduces `results/grid/gpt2_v2/` bit-identically, maximum absolute difference 0.000e+00 on all three recorded channels.

3.4 Result, framing 1 (full vocabulary), $n = 200$

Source: `results/ctg/studyA/A.fullvocab.*.json`, index `results/ctg/studyA_summary.json`.

Arm	Exact %	Final KL	Accept %	fwd-eq/seq	GPU s/seq	GPU s/accept
A1 <code>policy_self</code>	24.5	4.598	25.2	100	4.18	0.332
A2 <code>policy_onehot</code>	26.5	4.584	24.3	400	10.71	0.881
A4 <code>policy</code>	0.0	6.243	11.1	300	7.71	1.391
A5 <code>uniform</code>	0.0	6.671	9.9	100	4.24	0.857

Table 2: Study A, full vocabulary. Paired bootstrap contrasts against A1: A2 gives ΔKL -0.015 $[-0.361, +0.331]$ and Δexact +2.0 points $[-0.5, +5.0]$ at four times the forward-equivalent cost; A4 gives -24.5 points $[-30.5, -18.5]$; A5 gives -24.5 points $[-30.5, -18.5]$.

Prediction 1 is confirmed strongly. The entire linearized future term, the thing that costs the backward pass, buys no measurable quality. Put the way a defense question wants it: *the thesis’s recovery result is obtainable with no backward pass at all*. `policy_self` never calls `autograd` once, which the run’s own counter certifies, and it recovers 24.5% where the input-embedding gradient recovers 0.0% on the same two hundred sequences.

This also answers a question the submitted thesis explicitly leaves open. Section 5.6.1 observes that at the final position the future term vanishes so the token-indicator derivative reduces exactly to $\log p(v \mid \mathbf{x}_{<i})$, and adds that “in the interior the two come apart, because the future term is then nonzero”. Measured in the interior, they do not come apart to any degree the data can resolve.

k	Arm	Exact %	Final KL	Accept %	fwd-eq/seq	GPU s/seq
16	policy_self	30.0	4.788	30.4	100	4.22
16	policy_onehot	30.0	4.948	30.4	400	10.57
16	policy_exact_k	30.5	4.879	80.2	117	4.41
16	policy	30.5	4.814	17.0	400	10.42
16	uniform	31.5	4.580	16.6	100	5.56
64	policy_self	29.5	4.706	25.9	100	5.44
64	policy_onehot	28.0	4.822	25.7	400	10.63
64	policy_exact_k	31.0	4.150	73.0	165	4.68
64	policy	21.5	4.716	12.6	400	11.66
64	uniform	18.5	4.875	13.4	100	4.26
256	policy_self	26.5	4.638	23.0	100	4.24
256	policy_onehot	25.0	4.660	22.7	400	10.91
256	policy_exact_k	31.0	4.037	66.6	357	5.58
256	policy	8.5	5.271	11.1	400	11.42
256	uniform	8.0	5.195	12.4	100	4.22

Table 3: Study A, matched shortlist. All arms propose over the same frozen candidate set.

k	Contrast	ΔKL	Δexact (pts)	Cost
16	onehot – self	+0.160 [−0.011, +0.360]	+0.0 [−2.0, +2.0]	×4.00 fwd-eq
16	exact _k – self	+0.090 [−0.264, +0.442]	+0.5 [−3.0, +4.0]	×1.17 fwd-eq
64	onehot – self	+0.116 [−0.121, +0.363]	−1.5 [−4.0, +0.5]	×4.00 fwd-eq
64	exact_k – self	−0.555 [−0.977, −0.161]	+1.5 [−2.0, +5.5]	×1.65 fwd-eq, × 0.86 wall
256	onehot – self	+0.023 [−0.212, +0.269]	−1.5 [−4.0, +1.0]	×4.00 fwd-eq
256	exact_k – self	−0.601 [−1.008, −0.205]	+4.5 [+0.0, +9.0]	×3.57 fwd-eq

Table 4: Paired contrasts against the self term, 10,000-sample bootstrap, paired on `sample_idx`. The corruption is deterministic per index and identical across arms, so the pairing needs no rerun.

3.5 Result, framing 2 (matched shortlist), $n = 200$ per cell

Prediction 2 is refuted, and is reported as refuted. The exact future term does buy a real improvement once the shortlist is wide enough for ranking to matter: -0.555 nats at $k = 64$ and -0.601 at $k = 256$, both confidence intervals excluding zero, plus $+4.5$ exact-match points at $k = 256$. It is also cheap, which was the unexpected part: at $k = 64$ it costs 1.65 times the forward-equivalents and 0.86 times the wall-clock, that is, it is faster in real time than the arm it beats. Two measured reasons: its acceptance rate is 66 to 80% against 23 to 30% for every other arm, so it wastes far fewer steps; and with one masked position the k candidate energies do not depend on the current token, so they are computed once per sequence and reused. That second economy is task-specific and does not survive to a task where many positions move.

3.6 The support, not the scoring rule, is the dominant factor

Arm	Full vocab	$k = 256$	$k = 64$	$k = 16$
uniform (no scoring at all)	0.0	8.0	18.5	31.5
policy (input-embedding gradient)	0.0	8.5	21.5	30.5
policy_self	24.5	26.5	29.5	30.0
gradient minus uniform (points)	0.0	+0.5	+3.0	-1.0

Table 5: Exact recovery by shortlist width. At $k = 16$ every arm lands between 30.0 and 31.5 percent, including the arm that scores nothing.

This does *not* rehabilitate the input-embedding gradient. Its correct control is the uniform arm, and it tracks that floor at every width, never separating from it. What it does mean is that any claim of the form “surrogate X works in a shortlisted sampler” must be quoted against the uniform-on-the-same-shortlist floor rather than against zero. That floor was not measured anywhere in the submitted thesis, and holding the support fixed is a stronger statement of the null than the thesis makes, because it rules out the objection that the null was an artifact of proposing over 50,257 candidates.

4 A correction to a claim the submitted thesis makes

The submitted thesis, `chapters/05_results.tex` line 413, states that the token-indicator surrogate “recovers 36 to 40 percent of corrupted tokens”, and line 462 describes the outcome as “recovery comparable to a purpose-trained diffusion model”.

n	Exact recovery	95% CI
50 (the sweep cell as published)	40.0%	[26.4, 53.6]
200 (same configuration, rerun)	26.5%	[20.4, 32.6]

Table 6: The same cell at two sample sizes. Sources `results/grid/rev3/ohsweep_e10p5_t1p0.json` and `results/ctg/studyA/A.fullvocab.policy_onehot.json`.

The published number is reproducible and is not an error: rerunning the configuration and taking the same first fifty sequences gives exactly 40.0 percent. What the larger sample shows is that 40.0 percent was an $n = 50$ estimate which regresses to 26.5 percent at $n = 200$, below the range the text states.

The comparability claim is the part that does not survive. The comparators it is set against, SEDD-small 38.5%, SEDD-medium 39.0% and RoBERTa-large 44.5%, were all measured at

$n = 200$. The thesis therefore matches an $n = 50$ estimate against $n = 200$ estimates. At equal n the token-indicator sampler reaches 26.5% [20.4, 32.6] and is below both rather than comparable to either.

What survives unchanged, and it is the load-bearing part. The qualitative claim is not weakened and is now measured at four times the sample size: the token-indicator surrogate recovers 26.5 percent where the input-embedding gradient recovers 0.0 percent on the same sequences, a paired difference of 24.5 points, CI [18.5, 30.5]. The direction, the mechanism and the diagnosis are untouched. What changes is the size of the number and the claim of parity with a diffusion model.

5 The mucola_faithful arm

5.1 Why it exists

The thesis’s continuous sampler is a variant of MuCoLa, not MuCoLa. Following Kumar, Paria and Tsvetkov (2022), section 3, the MuCoLa language-model term is

$$\log P(\tilde{\mathbf{e}}_{n+1} \mid \tilde{\mathbf{e}}_{1:n}) = \mathbf{h}_n^\top \tilde{\mathbf{e}}_{n+1} + b_{n+1} - \log \sum_j \exp(\mathbf{h}_n^\top \mathbf{e}_j + b_j),$$

with the continuous state $\tilde{\mathbf{e}}$ in the softmax *numerator* and the real embedding table in the denominator. GPT-2’s output head is tied to the input embedding and carries no bias, so $b = 0$. The state therefore receives gradient by both paths the paper names: directly, with gradient exactly $\mathbf{h}_n$, and through the following hidden states.

The repository’s energy instead gathers the score at the projected token *index*, so the direct path is absent by construction. That is the object the thesis’s null characterizes.

5.2 Verification against the paper

Run before the arm was used for anything.

Check	Expected	Measured
MuCoLa energy at a real token sequence	equals the ordinary sequence log-likelihood, since $\tilde{\mathbf{e}} = \mathbf{e}(x)$ recovers the standard softmax	−46.909576 vs −46.909576, abs. diff. 0.000e+00
Gradient at the final position, MuCoLa energy	exactly $\mathbf{h}_{n-1}$, the paper’s direct path	norm 50.5592 vs $\ \mathbf{h}_{n-1}\ = 50.5592$, cosine 1.000000
Gradient at the final position, thesis energy	exactly zero (the thesis’s own theorem)	0.000e+00

Table 7: Verification of the MuCoLa energy implementation.

The third row reproduces the thesis’s final-position theorem inside this code, and the second row is precisely why that theorem does not apply to MuCoLa.

5.3 The two energies head to head

Identical code, optimizer, projection, initialization, step count and constraint weight. The only difference is how the target token enters the language-model term.

First generated continuation from each, same prompt and seed:

Variant	Steering-head acc.	Mean final LM log-prob
<code>mucola_faithful</code>	40.0%	−38.5
<code>cls_thesis</code>	53.3%	−353.9

Table 8: Smoke scale, $n = 15$, 60 steps. The steering-head column is deliberately included; see the warning below.

`mucola_faithful`: “Once upon a time, there was a king who ruled over a large land. His subjects were very happy, and he”

`cls_thesis`: “Once upon a time environmentalCNN Unicornrosso PE ##YD vitality ped Eh Karn bladesActionCodeStreamerBot Rothido Osborne”

MuCoLa’s energy produces fluent, coherent English. The energy this repository implements produces word salad at a sequence log-probability roughly an order of magnitude worse.

The inverted steering number, and why a held-out judge is mandatory. `cls_thesis` scores *higher* on the steering classifier while producing gibberish. A pooled sentiment head over hidden states is trivially satisfiable by incoherent token soup, so an adherence number reported without a fluency number is worthless. Every adherence figure in this programme is paired with an external fluency figure for exactly this reason, and the classifier that steers is never the classifier that scores.

5.4 This confirms the submitted thesis rather than contradicting it

`chapters/06_discussion.tex` line 50 already states that the thesis “refutes the premise for a sampler that differentiates the input embedding of a likelihood whose target token enters as a discrete index, and not for gradient-guided discrete sampling as such”, and describes MuCoLa’s self term correctly. The correction was applied before submission. This experiment turns an argument made from the papers’ text into a measurement: implementing MuCoLa’s energy in this repository’s own code makes the continuous sampler work, and implementing the repository’s energy makes it fail, on the same task with everything else held fixed.

6 Study C: how should the constraint enter?

6.1 Shortlist coverage, reported before any adherence number

The diagnostic: at each step, find the candidate the steering classifier most prefers over the whole vocabulary, then ask where it falls in the likelihood ranking and whether it lies inside the shortlist. Source `results/ctg/phase2/Ccov.k{16,64,256}.json`, $n = 30$ each.

k	Coverage	Chance level	Ratio	Median likelihood rank	Linearization ρ
16	0.2%	0.032%	$\sim 6\times$	29,928/50,257	+0.019
64	0.8%	0.127%	$\sim 6\times$	34,227/50,257	+0.040
256	1.7%	0.509%	$\sim 3\times$	29,866/50,257	−0.008

Table 9: Shortlist coverage and the constraint-side linearization correlation.

Finding 1: the shortlist almost never contains what the constraint wants. The median likelihood rank of the constraint’s preferred token is around 30,000 out of 50,257, the middle of the vocabulary: the two rankings are close to independent. Coverage is three to six times chance,

so the association is positive and real, but it is still under 2 percent even at $k = 256$. Widening the shortlist sixteen-fold moves coverage from 0.2 to 1.7 percent.

The consequence was pre-registered: any weakness of an exact-rescoring arm is a *proposal-support limitation*, not evidence about rescoring as a mechanism, because such an arm can only rescore candidates the shortlist contains.

Finding 2: the constraint-side linearization carries no usable signal. The rank correlation between the classifier’s first-order surrogate and the true change in the classifier’s score over the same candidates is +0.019, +0.040 and -0.008 , indistinguishable from zero.

Surrogate	Spearman against the true change
Likelihood, token-indicator, admissible distances	+0.60 to +0.73
Likelihood, input-embedding gradient	~ 0.0
Constraint, first-order classifier gradient	+0.02 to +0.06

Table 10: The classifier gradient is in the same condition as the input-embedding gradient the thesis’s null is about.

A qualification, not a contradiction. The repository records the concern-11 reading as “the constraint gradient’s direction carries signal, the LM gradient’s does not”, resting on a paired contrast of roughly +27 to +37 points on the continuation setup. That contrast is not re-run here and is not disputed. What is added is that whatever signal the constraint direction carries is not carried by the accuracy of its first-order expansion. A plausible reconciliation, offered as a hypothesis rather than a result: a classifier gradient can point usefully in the aggregate over many small continuous steps, which is what the continuous sampler takes, while being a poor predictor of the effect of any single discrete token substitution, which is what this measurement scores.

A tautological number, flagged so it is never misread. The mixture-shortlist arm reports 100.0% coverage by construction, because it builds half its shortlist from the constraint’s own ranking. It is reported for completeness and must never be presented as that arm achieving coverage.

7 Study A on the constrained-generation task

Partial at the time of writing; `exact_k` and `uniform` were still running. $n = 60$ sequences, span 20, 600 steps, $k = 64$, constraint in the accept step only.

Arm	Steer-head acc.	Held-out adherence	Accept %	fwd-eq/seq	GPU s
self	36.7%	$46.7\% \pm 12.6$	10.3	635	2,982
onehot	41.7%	$63.3\% \pm 12.2$	10.3	4,133	8,047
embgrad	38.3%	$56.7\% \pm 12.5$	12.9	4,234	10,712

Table 11: Held-out adherence is scored by an off-the-shelf sentiment classifier with no connection to this project.

The three adherence figures have 95% intervals of roughly ± 12.5 points on $n = 60$ and overlap heavily, so the apparent ordering is *not* a separation. The better powered paired test, on the same prompt and seed across arms, puts both differences on zero: `onehot` – `self`

gives +0.138 [−0.186, +0.449] on the steering log-probability, and `embgrad − self` gives +0.034 [−0.255, +0.320].

Consistent with the recovery result: the self term matches the full surrogate at 6.5 times less compute. Q1’s answer does not change when the task changes from repairing one token to conjuring twenty.

One result that goes the other way. `embgrad` attains a higher final LM log-probability, +12.52 nats with a CI excluding zero. This is not evidence of directional signal: every arm uses the identical exact accept step and therefore targets the identical distribution, so a difference in final energy is a difference in *mixing*, and `embgrad`’s acceptance rate is indeed higher. It is recorded rather than filtered out.

8 Incidents, defects and harness lessons

8.1 Out-of-memory in the nearest-embedding projection

Six jobs failed with CUDA out-of-memory at `core/prop.py` line 41. The shared projection helper forms the difference tensor explicitly, of shape (M, V, D) . At one masked position that is 0.26 GB, which is why the entire 145-configuration recovery grid never touched it; at the twenty positions of the continuation task it is 4.79 GB in a single allocation, every step.

The fix is a local low-memory projection using $\|\mathbf{s} - \mathbf{e}\|^2 = \|\mathbf{s}\|^2 + \|\mathbf{e}\|^2 - 2\mathbf{s}^\top \mathbf{e}$ and dropping the term constant in the candidate. `core/prop.py` is deliberately not changed, because it is on the archived path. Equivalence was verified on CPU over 120 projections in three regimes: *zero mismatches*.

8.2 A constraint-weight sweep that ran backwards

The continuous-optimizer arms initially showed the steering score getting *worse* as the constraint weight increased, $-1.240 \rightarrow -1.268 \rightarrow -1.614$ at gains 0.5, 2.0, 4.0, with the generated text identical across gains. The cause is plain gradient ascent without normalization: the gradient norm scales with the weights, so the step length scales with the constraint weight and the iterate overshoots at the schedule’s initial step size, projecting to unrelated tokens.

The fix is to unit-norm the combined gradient per position before stepping. This leaves the direction, and hence the weighting between the two terms, untouched, and makes the step length independent of the weight. With it the sweep behaves monotonically (73.3% $\rightarrow$ 80.0% $\rightarrow$ 80.0% at gains 0.5, 4.0, 8.0). This matters for fairness: an unnormalized sweep would have understated MuCoLa’s arm, which is the arm this programme is at pains to represent fairly.

8.3 Lessons that generalize

- A single-mask diagnostic task hides a class of scaling bug. Two separate defects here were invisible at one masked position and appeared immediately at twenty.
- Reversibility failures are silent in the worst way: acceptance collapses to exactly zero while the proposal itself is perfectly healthy. The diagnosis required measuring the proposal mass and the energy gain of the refused move.
- The manifest is a priority list and can be reordered live; workers re-read it each pass. A gating experiment stuck behind a long family is a scheduling problem, not a compute problem.
- Worker counts should be set by measurement. Two workers per card left the GPUs at 15 to 43 percent utilization on these chains; four per card reached 100 percent.

9 What was not done, stated rather than omitted

- **Study B** (proposal sources including RoBERTa and SEDD on the constrained task) was gated on the Study C mechanism probe and was not launched. The harness supports both proposals and both were validated in the recovery setting by the earlier revision work.
- **Study C’s full mechanism comparison** at scale was in flight; only the coverage half is reported above as complete.
- **Study D’s** full adherence-fluency curve at matched compute was in flight.
- **Study E**, diversity at matched adherence against a compute-matched best-of- N optimizer, was not reached. The metrics are implemented and validated (distinct-1 0.929, distinct-3 1.00, self-BLEU 0.000, semantic spread 0.778 on one arm) but the controlled comparison is the part that matters, so no claim about posterior coverage is made on this evidence.

10 Summary for the defense

1. The constructive half of the thesis needs no gradient at all. The self term, which every autoregressive model already exposes, gets the whole effect at a quarter of the cost and zero backward passes.
2. The 40 percent headline is an $n = 50$ cell that reproduces exactly; at $n = 200$ it is 26.5 percent, so the honest claim is zero to roughly a quarter recovered, below SEDD and RoBERTa rather than at parity with them. The zero-versus-nonzero contrast, which is what the thesis rests on, is unaffected and better supported.
3. The scope restriction the thesis states about MuCoLa is correct, and can now be demonstrated rather than argued: the same code with MuCoLa’s energy writes fluent English and with the thesis’s energy writes word salad.
4. A shortlisted sampler needs a uniform-on-the-same-shortlist floor before any surrogate claim can be read, and that floor was never measured in the thesis.
5. Gradient-based constraint guidance is weak here for a measurable reason: the classifier’s first-order expansion has a rank correlation near zero with the change it is expanding, and a likelihood-ranked shortlist almost never contains the token the constraint wants.